

Introduction to Word Embeddings

- Goal: represent word meaning in a form computers can use
- Problem: words are symbolic, but ML models need numbers
- Today: one-hot vectors, distributional semantics, dense embeddings

One-hot vectors

- Each word is represented as a long sparse vector
- Only one position is 1; all others are 0
- Limitation: no natural similarity between related words

Distributional hypothesis

- Words that occur in similar contexts often have similar meanings
- Context provides evidence about meaning
- Example: cat and dog appear near words like pet, food, animal

Dense embeddings

- Embeddings represent words as shorter dense vectors
- Similar words should have vectors close together
- Cosine similarity is commonly used to compare vectors

Recap

- One-hot vectors are simple but sparse
- Embeddings capture similarity using context
- Useful for classification, search, translation, and other NLP tasks